

**BICAL-50**  
(Bicalutamide Tablets 50 mg)

**Name and strength of active ingredient**

Bicalutamide 50 mg

**Dosage form**

Film Coated Tablet

**Therapeutic Class**

Antineoplastic and Immunomodulating Agents (Anti-Androgens)

**Product Description**

White to off white, round, biconvex, film-coated tablets debossed 'B 50' on one side and plain on other side.

**Pharmacological properties**

**Pharmacodynamic properties**

Pharmacotherapeutic group: Anti-androgens, ATC code L02BB03

Mechanism of action

Bicalutamide Tablets 50 mg is a non-steroidal anti-androgen, devoid of other endocrine activity. It binds to androgen receptors without activating gene expression, and thus inhibits the androgen stimulus. Regression of prostatic tumours results from this inhibition. Clinically, discontinuation of Bicalutamide Tablets 50 mg can result in antiandrogen withdrawal syndrome in a subset of patients.

Bicalutamide Tablets 50 mg is a racemate with its antiandrogenic activity being almost exclusively in the (R)-enantiomer.

**Pharmacokinetic properties**

Absorption

Bicalutamide Tablets 50 mg is well absorbed following oral administration. There is no evidence of any clinically relevant effect of food on bioavailability.

Distribution

Bicalutamide Tablets 50 mg is highly protein bound (racemate 96% (R)-enantiomer >99%) and extensively metabolised (via oxidation and glucuronidation): Its metabolites are eliminated via the kidneys and bile in approximately equal proportions.

Biotransformation

The (S)-enantiomer is rapidly cleared relative to the (R)-enantiomer, the latter having a plasma elimination half-life of about 1 week.

On daily administration of Bicalutamide Tablets 50 mg, the (R)-enantiomer accumulates about 10-fold in plasma as a consequence of its long half-life.

Steady state plasma concentrations of the (R)-enantiomer of approximately 9 microgram/ ml are observed during daily administration of 50 mg doses of Bicalutamide Tablets 50 mg. At steady state the predominantly active (R)-enantiomer accounts for 99% of the total circulating enantiomers.

Elimination

In a clinical study the mean concentration of R-bicalutamide in semen of men receiving bicalutamide 150 mg was 4.9 microgram/ml. The amount of bicalutamide potentially delivered to a female partner during intercourse is low and by extrapolation possibly equates to approximately 0.3 microgram/kg. This is below that required to induce changes in offspring of laboratory animals.

Special Populations

The pharmacokinetics of the (R)-enantiomer are unaffected by age, renal impairment or mild to moderate hepatic impairment. There is evidence that for subjects with severe hepatic impairment, the (R)-enantiomer is more slowly eliminated from plasma.

**Preclinical safety data**

Bicalutamide is a potent antiandrogen and a mixed function oxidase enzyme inducer in animals. Target organ changes, including tumour induction, in animals, are related to these activities. Atrophy of seminiferous tubules of the testes is a predicted class effect with anti-androgens and has been observed for all species examined. Reversal of testicular atrophy occurred 4 months after the completion of dosing in a 6-month rat study (at doses of approximately 1.5 times human therapeutic concentrations at the recommended dose of 50 mg). No recovery was observed at 24 weeks after the completion of dosing in a 12-month rat study (at doses of approximately 2 times human concentrations at the recommended human dose of 50 mg). Following 12-months of repeated dosing in dogs (at doses of approximately 7 times human therapeutic concentrations at the recommended human dose of 50 mg), the incidence of testicular atrophy was the same in dosed and control dogs after a 6-month recovery period. In a fertility study (at doses of approximately 1.5 times human therapeutic concentrations at the recommended human dose of 50 mg), male rats had an increased time to successful mating immediately after 11 weeks of dosing; reversal was observed after 7 weeks off-dose.

**Indication/Usage**

Treatment of advanced prostate cancer in combination with LHRH analogue therapy or surgical castration.

**Recommended Dose**

Adult : Males including the elderly:

One tablet (50 mg) once a day.

Treatment with Bicalutamide Tablets 50 mg should be started at the same time as treatment with an LHRH analogue or surgical castration.

Children:

Bicalutamide Tablets 50 mg is contra-indicated in children.

Renal impairment:

No dosage adjustment is necessary for patients with renal impairment.

Hepatic impairment:

No dosage adjustment is necessary for patients with mild hepatic impairment. Increased accumulation may occur in patients with moderate to severe hepatic impairment.

**Route of Administration**

Oral

**Contraindication**

Bicalutamide Tablets 50 mg is contra-indicated in females and children. Bicalutamide Tablets 50 mg must not be given to any patient who has shown a hypersensitivity reaction to its use. Coadministration of terfenadine, astemizole or cisapride with Bicalutamide Tablets 50 mg is contraindicated.

**Warnings & Precaution**

Initiation of treatment should be under the direct supervision of a specialist.

Bicalutamide Tablets 50 mg is extensively metabolised in the liver. Data suggests that its elimination may be slower in subjects with severe hepatic impairment and this could lead to increased accumulation of Bicalutamide Tablets 50 mg. Therefore, Bicalutamide Tablets 50 mg should be used with caution in patients with moderate to severe hepatic impairment.

Periodic liver function testing should be considered due to the possibility of hepatic changes. The majority of changes are expected to occur within the first 6 months of Bicalutamide Tablets 50 mg therapy.

Severe hepatic changes and hepatic failure have been observed rarely with Bicalutamide Tablets 50 mg, and fatal outcomes have been reported (see section 'Undesirable effects'). Bicalutamide Tablets 50 mg therapy should be discontinued if changes are severe.

A reduction in glucose tolerance has been observed in males receiving LHRH agonists. This may manifest as diabetes or loss of glycaemic control in those with pre-existing diabetes. Consideration should therefore be given to monitoring blood glucose in patients receiving Bicalutamide Tablets 50 mg in combination with LHRH agonists.

Bicalutamide Tablets 50 mg has been shown to inhibit Cytochrome P450 (CYP 3A4), as such caution should be exercised when co-administered with drugs metabolised predominantly by CYP 3A4 (see sections 'Contraindications' and 'Interaction with other medicinal products and other forms of interaction').

Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Androgen deprivation therapy may prolong the QT interval.

In patients with a history of or risk factors for QT prolongation and in patients receiving concomitant medicinal products that might prolong the QT interval (see section 'Interaction with other medicinal products and other forms of interaction') physicians should assess the benefit risk ratio including the potential for Torsade de pointes prior to initiating Bicalutamide Tablets 50 mg.

Antiandrogen therapy may cause morphological changes in spermatozoa. Although the effect of bicalutamide on sperm morphology has not been evaluated and no such changes have been reported for patients who received Bicalutamide Tablets 50 mg, patients and/or their partners should follow adequate contraception during and for 130 days after Bicalutamide Tablets 50 mg therapy.

Potentiation of coumarin anticoagulant effects have been reported in patients receiving concomitant Bicalutamide Tablets 50 mg therapy, which may result in increased Prothrombin Time (PT) and International Normalised Ratio (INR). Some cases have been associated with risk of bleeding. Close monitoring of PT/INR is advised and anticoagulant dose adjustment should be considered (see sections 'Interaction with other medicinal products and other forms of interaction' and 'Undesirable effects').

**Interaction with other medications**

There is no evidence of any pharmacodynamic or pharmacokinetic interactions between Bicalutamide Tablets 50 mg and LHRH analogues. In vitro studies have shown that R-bicalutamide is an inhibitor of CYP 3A4, with lesser inhibitory effects on CYP 2C9, 2C19 and 2D6 activity.

Although clinical studies using antipyrine as a marker of cytochrome P450 (CYP) activity showed no evidence of a drug interaction potential with ' Bicalutamide Tablets 50 mg, mean midazolam exposure (AUC) was increased by up to 80%, after co-administration of Bicalutamide Tablets 50 mg for 28 days. For drugs with a narrow therapeutic index such an increase could be of relevance. As such, concomitant use of terfenadine, astemizole and cisapride is contra-indicated and caution should be exercised with the co-administration of Bicalutamide Tablets 50 mg with compounds such as cyclosporin and calcium channel blockers. Dosage reduction may be required for these drugs particularly if there is evidence of enhanced or adverse drug effect. For cyclosporin, it is recommended that plasma concentrations and clinical condition are closely monitored following initiation or cessation of Bicalutamide Tablets 50 mg therapy. Caution should be exercised when prescribing Bicalutamide Tablets 50 mg with other drugs which may inhibit drug oxidation e.g. cimetidine and ketoconazole. In theory, this could result in increased plasma concentrations of Bicalutamide Tablets 50 mg, which theoretically could lead to an increase in side effects. In vitro studies have shown that Bicalutamide Tablets 50 mg can displace the coumarin anticoagulant, warfarin, from its protein binding sites. It is therefore recommended that if Bicalutamide Tablets 50 mg is started in patients who are already receiving coumarin anticoagulants, prothrombin time should be closely monitored.

**Pregnancy and Lactation**

Bicalutamide Tablets 50 mg is contra-indicated in females and must not be given to pregnant women or nursing mothers.

**Side effects/Adverse Reactions**

Bicalutamide in general, has been well tolerated with few withdrawals due to adverse events.

**Table 1 Frequency of Adverse Reactions**

| Frequency                   | System Organ Class                                  | Event                                                                                                                           |
|-----------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Very common (1/10)          | Reproductive system and breast disorders            | Breast tenderness <sup>1</sup>                                                                                                  |
|                             |                                                     | Gynaecomastia <sup>1</sup>                                                                                                      |
|                             |                                                     | Decreased libido, erectile dysfunction, impotence                                                                               |
| Common (1/100, <1/10)       | General disorders                                   | Hot flushes <sup>1</sup>                                                                                                        |
|                             | Gastrointestinal disorders                          | Diarrhoea                                                                                                                       |
|                             |                                                     | Nausea                                                                                                                          |
|                             |                                                     | Constipation                                                                                                                    |
|                             | Hepato-biliary disorders                            | Hepatic changes (elevated levels of transaminases, cholestasis and jaundice) <sup>2</sup>                                       |
|                             | General disorders and administrative conditions     | Asthenia                                                                                                                        |
|                             |                                                     | Pruritus                                                                                                                        |
|                             |                                                     | Oedema, General pain, pelvic pain, chills                                                                                       |
|                             | Blood and Lymphatic system disorders                | Anaemia                                                                                                                         |
|                             | Metabolism and nutrition disorders                  | Diabetes mellitus, weight gain.                                                                                                 |
| Uncommon (1/1,000, < 1/100) | Nervous system disorders                            | Dizziness, insomnia.                                                                                                            |
|                             | Skin and subcutaneous tissue disorders              | Rash, sweating, hirsutism                                                                                                       |
|                             | Immune system disorders                             | Hypersensitivity reactions, including angioneurotic oedema and urticaria                                                        |
|                             | Respiratory, thoracic and mediastinal disorders     | Interstitial lung disease, dyspnoea                                                                                             |
|                             | Metabolism and nutrition disorders                  | Anorexia, hyperglycaemia, Weight loss.                                                                                          |
|                             | Nervous system disorders                            | Somnolence                                                                                                                      |
|                             | Skin and subcutaneous tissue disorders              | Alopecia                                                                                                                        |
|                             | Renal and urinary disorders                         | Nocturia                                                                                                                        |
|                             | General disorders and administration site condition | Abdominal pain, chest pain, headache, pain in the back, neck pain.                                                              |
|                             | Gastrointestinal disorders                          | Dry mouth, dyspepsia, flatulence.                                                                                               |
| Rare (1/10,000, <1/1,000)   | Gastrointestinal disorders                          | Vomiting                                                                                                                        |
|                             | Skin and subcutaneous tissue disorders              | Dry skin                                                                                                                        |
|                             | Hepato- biliary disorders                           | Hepatic failure <sup>3</sup>                                                                                                    |
| Very rare (<1/10,000)       | Blood and lymphatic system disorders                | Thrombocytopenia has been reported rarely.                                                                                      |
|                             | Cardiac disorders                                   | Angina, heart failure, conduction defects including PR and QT interval prolongations, arrhythmias and non-specific ECG changes. |

1. May be reduced by concomitant castration.
2. Hepatic changes are rarely severe and were frequently transient, resolving or improving with continued therapy or following cessation of therapy.
- 3.Hepatic failure has occurred very rarely in patients treated with Bicalutamide but a causal relationship has not been established with certainty. Periodic liver function testing should be considered.

**Signs & Symptoms of overdose and Treatment**

No case of overdose has been reported. There is no specific antidote; treatment should be symptomatic. Dialysis may not be helpful, since Bicalutamide is highly protein bound and is not recovered unchanged in the urine. General supportive care, including frequent monitoring of vital signs, is indicated.

**Storage**

Store below 30°C.

**Shelf life**

24 months from date of manufacturing.

**Pack size**

PVC/PVdC Alu Blister of 10 tablets. Each Carton contains 3 such blister. (3 x 10T)

Manufactured by:



**INTAS PHARMACEUTICALS LTD.**

Matoda-382 210, Dist. : Ahmedabad. INDIA

Product Registration Holder:

**Jetpharma Sdn Bhd**

No.13, Jalan Rajawali 2, Bandar Puchong Jaya

47100 Puchong, Selangor, Malaysia

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Front Side

Back Side

**File Name : BICAL-50(MALAYSIA)PIL**  
**Size : 175 x 275 (mm)**  
**Colour : Pantone Black**  
**Date : 31/10/22, 15/11/22, 17/11/22, 27/04/23**